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ANTI-THEFT BRACKET AND VEHICLE CONTROL DEVICE INCLUDING THE SAME

Abstract

An anti-theft bracket and a vehicle control device therefor are provided. The anti-theft bracket for the vehicle control device includes a connector withdrawal prevention portion configured to block an insertion connector fitted with a housing of the vehicle control device from being withdrawn from the housing, and a housing coupling portion connected to the connector withdrawal prevention portion and fitted with the housing. The housing coupling portion prevents the housing coupling portion and the connector withdrawal prevention portion from being withdrawn from the housing.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application claims priority from and the benefit under 35 USC § 119 of Korean Patent Application No. 10-2024-0030298, filed on Feb. 29, 2024, in the Korean Intellectual Property Office, the entire disclosure of which is incorporated by reference for all purposes.

BACKGROUND

1. Field

[0002] Exemplary embodiments of the present disclosure relate to an anti-theft bracket, and more particularly, to an anti-theft bracket coupled to a vehicle control device, and the vehicle control device including the same.

2. Description of the Related Art

[0003] Canada and some European countries, for example, have regulations in place to suppress criminal activity that involves removing connectors from vehicle control devices, such as body control modules (BCMs) and integrated body units (IBUs), and connecting cables to hack or steal vehicles. Therefore, there is a need for an anti-theft bracket that meets the requirement of the regulation.

[0004] A background technology related to the present disclosure is disclosed in Korean Patent Application Laid-Open No. 10-2019-0056384 (published on May 24, 2019 and entitled ‘Vehicle Anti-Theft Device’).

SUMMARY

[0005] This Summary is provided to introduce a selection of concepts in simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

[0006] In a general aspect of the disclosure, an anti-theft bracket for a vehicle control device, includes: a connector withdrawal prevention portion configured to block an insertion connector fitted with a housing of the vehicle control device from being withdrawn from the housing; and a housing coupling portion connected to the connector withdrawal prevention portion and fitted with the housing, wherein the housing coupling portion is configured to prevent the housing coupling portion and the connector withdrawal prevention portion from being withdrawn from the housing.

[0007] The vehicle control device may include at least one of a body control module (BCM), an integrated body unit (IBU), an engine control module (ECM), or any combination thereof.

[0008] The housing of the vehicle control device may include a circuit board installed therein, and the insertion connector may be electrically coupled to the circuit board.

[0009] In another general aspect of the disclosure, a vehicle control device includes: a housing including: a connector fitting part; and a bracket fitting part provided at one side of the connector fitting part; an insertion connector fitted with the connector fitting part; and an anti-theft bracket including: a connector withdrawal prevention portion configured to block the insertion connector fitted with the housing from being withdrawn from the connector fitting part; and a housing coupling portion connected to the connector withdrawal prevention portion and fitted with the bracket fitting part, wherein the housing coupling portion is configured to prevent the housing coupling portion from being withdrawn from the bracket fitting part.

[0010] The housing coupling portion may include a bracket hook protruding in a direction intersecting a direction in which the anti-theft bracket is inserted into the bracket fitting part.

[0011] The housing may include a bracket hook stopper configured to block the bracket hook so that the housing coupling portion inserted into the bracket fitting part does not move in a direction opposite to a direction in which the housing coupling portion is inserted into the bracket fitting part.

[0012] The housing may include a housing hook protruding in a direction intersecting a direction in which the anti-theft bracket is inserted into the bracket fitting part.

[0013] The housing coupling portion may include a housing hook stopper blocked by the housing hook so that the housing coupling portion inserted into the bracket fitting part does not move in a direction opposite to a direction in which the housing coupling portion is inserted into the bracket fitting part.

[0014] The connector withdrawal prevention portion may cover an inlet of the bracket fitting part.

[0015] The anti-theft bracket may further include a connector insertion portion connected to the connector withdrawal prevention portion and inserted into the insertion connector.

[0016] The insertion connector may include: a groove portion into which the connector insertion portion is inserted; and a bracket stopper part positioned between the housing coupling portion and the connector insertion portion and facing the connector withdrawal prevention portion when the housing coupling portion is fitted with the bracket fitting part.

[0017] The anti-theft bracket may further include a blade portion being in surface contact with an inner surface that defines an inlet of the bracket fitting part.

[0018] The blade portion may be bent and extend from the housing coupling portion, and the connector withdrawal prevention portion may block the blade portion so that the blade portion is not folded in a direction in which an angle between the blade portion and the housing coupling portion decreases.

[0019] The anti-theft bracket may be manufactured by performing a press processing on a metal board.

[0020] The vehicle control device may include at least one of a body control module (BCM), an integrated body unit (IBU), an engine control module (ECM), or any combination thereof.

[0021] The housing of the vehicle control device may include a circuit board installed therein, and the insertion connector may be electrically coupled to the circuit board.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] FIG. 1 is a perspective view of a vehicle control device according to an embodiment of the present disclosure.

[0023] FIG. 2 is an enlarged perspective view of part II in FIG. 1, i.e., a view illustrating a state in which an insertion connector and an anti-theft bracket are spaced apart from a housing.

[0024] FIG. 3 is a cross-sectional view taken along line III-III in FIG. 1.

[0025] FIG. 4 is a cross-sectional view taken along line IV-IV in FIG. 1.

[0026] FIG. 5 is a partially enlarged front view illustrating a part of the housing having a connector fitting part and a bracket fitting part in FIG. 2.

[0027] FIG. 6 is a partially enlarged perspective view illustrating a cover illustrated in FIG. 5 when viewed from below.

[0028] FIG. 7 is a partially cut-away perspective view of the cover in FIG. 5 taken along line VII-VII when viewed from below.

[0029] FIG. 8 is a perspective view illustrating a state in which the insertion connector and the anti-theft bracket in FIG. 2 are separated when viewed from above.

[0030] FIG. 9 is a perspective view illustrating a state in which the insertion connector and the anti-theft bracket in FIG. 2 are separated when viewed from below.

DETAILED DESCRIPTION

[0031] Hereinafter, an anti-theft bracket and a vehicle control device including the same will be described in detail with reference to the accompanying drawings through various exemplary embodiments. The terminologies used in the present specification are used to appropriately express an exemplary embodiment of the present disclosure and may vary depending on the intention of a user or operator or on usual practice in the art to which the present disclosure pertains. Therefore, the definition of the terms should be made on the basis of the entire contents of the present specification.

[0032] FIG. 1 is a perspective view of a vehicle control device according to an embodiment of the present disclosure, FIG. 2 is an enlarged perspective view of part II in FIG. 1, i.e., a view illustrating a state in which an insertion connector and an anti-theft bracket are spaced apart from a housing, FIG. 3 is a cross-sectional view taken along line III-III in FIG. 1, FIG. 4 is a cross-sectional view taken along line IV-IV in FIG. 1, FIG. 5 is a partially enlarged front view illustrating a part of the housing having a connector fitting part and a bracket fitting part in FIG. 2, FIG. 6 is a partially enlarged perspective view illustrating a cover illustrated in FIG. 5 when viewed from below, FIG. 7 is a partially cut-away perspective view of the cover in FIG. 5 taken along line VII-VII when viewed from below, FIG. 8 is a perspective view illustrating a state in which the insertion connector and the anti-theft bracket in FIG. 2 are separated when viewed from above, and FIG. 9 is a perspective view illustrating a state in which the insertion connector and the anti-theft bracket in FIG. 2 are separated when viewed from below.

[0033] With reference to FIGS. 1 to 9, a vehicle control device **100** according to the embodiment of the present disclosure refers to a device configured to communicate with and control devices such as headlamps, power seats, an air conditioning device, a smart key, a driving assistance device, and a parking assistance device of a vehicle. For example, a body control module (BCM) or an integrated body unit (IBU) may be the vehicle control device **100**. In another embodiment of the present disclosure, the vehicle control device **100** may include a body control module (BCM), an integrated body unit (IBU), an engine control module (ECM), or any combination thereof.

[0034] The vehicle control device **100** may include a housing **101**, an insertion connector **160**, and an anti-theft bracket **200**. The housing **101** may include a base **102**, a cover **120**, and a fixing connector **110**. The cover **120** may be coupled to the base **102**. A circuit pattern may be formed in an internal space of the housing **101** defined by the base **102** and the cover **120**, and a circuit board on which electronic elements are mounted may be installed in the internal space of the housing **101**.

[0035] The fixing connector **110** may be installed at one side the base **102** and the cover **120** coupled to each other. A through-hole is formed at one side of the base **102** and the cover **120** and has a first inner interval **W1** and a second inner interval **W2** smaller than the first inner interval **W1**. In this case, among the inner surfaces that define a shape of the through-hole, an interval between a pair of first inner surfaces **124** facing each other is referred to as the first inner interval **W1**, and an interval between a pair of second inner surfaces **132** facing each other and stepped from the pair of first inner surfaces **124** is referred to as the second inner interval **W2**.

[0036] An outer surface of the fixing connector **110** is in surface contact with the pair of first inner surfaces **124**. The fixing connector **110** may be electrically coupled to the circuit board installed in the internal space of the housing **101**. More specifically, the fixing connector **110** may be electrically coupled to the circuit board fixedly supported on the base **102**, and the outer surface of the fixing connector **110** is in surface contact with and supported on the pair of first inner surfaces **124**, such that the fixing connector **110** may be fixedly supported on the base **102** and the cover **120**.

[0037] The housing **101** includes a connector fitting part **122** and a bracket fitting part **130**. The connector fitting part **122** may include a concave groove formed in the fixing connector **110** and formed to be visible from the outside of the housing **101**. The bracket fitting part **130** is provided at

one side of the connector fitting part **122**. For example, the bracket fitting part **130** may be provided above the connector fitting part **122**.

[0038] The insertion connector **160** is fitted with the connector fitting part **122**. More specifically, the insertion connector **160** may be fitted with the concave groove of the fixing connector **110**. The insertion connector **160** may include a connector body **161** and a connector cable **169**. The connector body **161** may be fitted with and fixed to the connector fitting part **122** to close the connector fitting part **122**.

[0039] When the connector body **161** is fitted with the connector fitting part **122**, a conductor of the fixing connector **110** and a conductor of the insertion connector **160** may be electrically connected. The connector cable **169** may be electrically connected to the conductor of the insertion connector **160** so that a device or sensor included in the vehicle and the circuit board installed in the housing **101** are connected in a communicable manner.

[0040] The anti-theft bracket **200** includes a connector withdrawal prevention portion **201** and a housing coupling portion **210**. The connector withdrawal prevention portion **201** blocks the insertion connector **160** to prevent the insertion connector **160** from being withdrawn from the connector fitting part **122**. The housing coupling portion **210** is connected to the connector withdrawal prevention portion **201** and fitted with the bracket fitting part **130** so as not to be withdrawn from the bracket fitting part **130**.

[0041] When the housing coupling portion **210** is fitted with the bracket fitting part **130** and fixed to the housing **101** so as not to be separated from the housing **101**, the connector withdrawal prevention portion **201** may cover an inlet of the bracket fitting part **130** so that the inlet of the bracket fitting part **130** is not exposed.

[0042] The cover **120** includes a bracket coupling plate **140** provided in the inlet of the bracket fitting part **130**. The bracket coupling plate **140** is a plate-shaped part extending in a leftward/rightward direction. Two opposite ends of the bracket coupling plate **140** are supported on the pair of second inner surfaces **132** facing each other. The housing coupling portion **210** enters between inner surfaces **133** of the cover **120** that face an upper surface of the bracket coupling plate **140** and an upper surface of the bracket coupling plate **140**, such that the housing coupling portion **210** is fitted with the bracket fitting part **130**.

[0043] The housing coupling portion **210** may include a bracket hook **211** protruding in a direction intersecting a direction in which the anti-theft bracket **200** is inserted into the bracket fitting part **130**. The bracket hook **211** may protrude to be inclined so as not to prevent the housing coupling portion **210** from entering the bracket fitting part **130**.

[0044] The housing **101** may include a bracket hook stopper **143** configured to block the bracket hook **211** so that the housing coupling portion **210** inserted into the bracket fitting part **130** does not move in a direction opposite to the direction in which the housing coupling portion **210** is inserted into the bracket fitting part **130**. The bracket hook stopper **143** may be provided at an end of the bracket coupling plate **140** in an inward direction of the housing **101**.

[0045] The bracket coupling plate **140** may further include an inclined surface **144** facing away from the bracket hook stopper **143**. When the housing coupling portion **210** is inserted into the housing **101** through the inlet of the bracket fitting part **130**, the inclined portion of the bracket hook **211** comes into contact with the inclined surface **144** of the bracket coupling plate **140**, and an inclination angle of the bracket hook **211** decreases, such that the bracket hook **211** may climb over the bracket hook stopper **143**.

[0046] An inclination angle of the bracket hook **211**, which has climbed over the bracket hook stopper **143**, is elastically restored to an original angle, such that the bracket hook **211** is blocked by the bracket hook stopper **143**. Therefore, the housing coupling portion **210** and the anti-theft bracket **200** including the same are not withdrawn from the bracket fitting part **130** as long as the housing coupling portion **210** and the anti-theft bracket **200** are damaged.

[0047] The housing **101** may include a housing hook **134** protruding in a direction intersecting the

direction in which the anti-theft bracket **200** is inserted into the bracket fitting part **130**. For example, the housing hook **134** may protrude from the inner surface **133** of the cover **120** that faces the upper surface of the bracket coupling plate **140**. In the embodiment of the present disclosure illustrated in FIGS. **2**, **5**, **8**, and **9**, the pair of housing hooks **134** is provided. However, another embodiment of the present disclosure may include three or more housing hooks or a single housing hook.

[0048] The housing coupling portion **210** may include a housing hook stopper **216** blocked by the housing hook so that the housing coupling portion **210** inserted into the bracket fitting part **130** does not move in the direction opposite to the direction in which the housing coupling portion **210** is inserted into the bracket fitting part **130**. For example, the housing hook stopper **216** may be provided as a pair of housing hook stoppers **216** provided at left and right sides of the bracket hook **211**. More specifically, a pair of housing hook accommodation portions **214**, which is recessed to accommodate the housing hooks **134**, may be formed at left and right sides of the bracket hook **211**, and the pair of housing hook stoppers **216** may be provided on the inner surface that defines the pair of housing hook accommodation portions **214**.

[0049] The housing hook **134** may include an inclined surface **135** inclined so that a distal end of the housing coupling portion **210** may climb over the housing hook **134** without being caught when the housing coupling portion **210** is inserted into the housing **101** through the inlet of the bracket fitting part **130**.

[0050] The bracket coupling plate **140** may further include inclined guides **142**, and protruding ribs **141**. The inclined guide **142** guides the housing coupling portion **130** so that the housing coupling portion becomes close to the inner surface **133** of the cover **120**, which faces the upper surface of the bracket coupling plate **140**, when the housing coupling portion **130** enters the bracket fitting part **130** through the inlet of the bracket fitting part **130**. The inclined guide **142** may be provided as a plurality of inclined guides **142** provided to avoid the bracket hook stopper **143**.

[0051] The protruding rib **141** may protrude downward so as to be in contact with the upper surface of the fixing connector **110**. The protruding rib **141** may be provided as a plurality of protruding ribs **141** spaced apart from one another. Because the bracket coupling plate **140** is supported on the fixing connector **110** by means of the protruding ribs **141**, the bracket coupling plate **140** may more securely support the housing coupling portion **210** and the anti-theft bracket **200** including the same without sagging downward.

[0052] The anti-theft bracket **200** may further include a connector insertion portion **220** and a blade portion **230**. The connector insertion portion **220** is connected to the connector withdrawal prevention portion **201** and inserted into the insertion connector **160**. The connector insertion portion **220** may be disposed below and spaced apart from the housing coupling portion **210**. The housing coupling portion **210** and the connector insertion portion **220** may extend in parallel with the direction in which the housing coupling portion **210** is inserted into the bracket fitting part **130**.

[0053] The insertion connector **160** includes a groove portion **167** and a bracket stopper part **165**. The connector insertion portion **220** may be inserted into the groove portion **167** when the housing coupling portion **210** is inserted into the bracket fitting part **130**. When the housing coupling portion **210** is fitted with the bracket fitting part **130**, the bracket stopper part **165** may be positioned between the housing coupling portion **210** and the connector insertion portion **220** and face the connector withdrawal prevention portion **201**.

[0054] The blade portion **230** is in surface contact with the second inner surface **132** that defines the inlet of the bracket fitting part **130**. For example, the blade portion **230** may be provided as a pair of blade portions **230** disposed one by one at left and right sides in the width direction of the connector withdrawal prevention portion **201**. The blade portion **230** may be bent and extend downward from the housing coupling portion **210**.

[0055] When the housing coupling portion **210** is fitted with the bracket fitting part **130**, an outer surface **232** of one side end of the blade portion **230** is in surface contact with the second inner

surface **132** in the longitudinal direction of the housing coupling portion **210**.

[0056] Therefore, in the state in which the insertion connector **160** is inserted and coupled into the connector fitting part **122** and the housing coupling portion **230** is inserted and coupled into the bracket fitting part **130**, no gap, through which a tool, such as a screwdriver or a lever, may be inserted is provided between the inlet of the bracket fitting part **130** and the anti-theft bracket **200**. Therefore, the anti-theft bracket **200** and the insertion connector **160** cannot be forcibly separated from the housing **10** within a short time. Therefore, the reliability of the function of preventing vehicle theft is improved.

[0057] The blade portion **230** may further include a stepped surface **236** that comes into contact with an edge of the fixing connector **110** when the housing coupling portion **210** is inserted into the bracket fitting part **130**. Therefore, in the state in which the insertion connector **160** is inserted and coupled into the connector fitting part **122** and the housing coupling portion **230** is inserted and coupled into the bracket fitting part **130**, no gap, into which a tool is inserted, is provided between the inlet of the bracket fitting part **130** and the inlet of the connector fitting part **122**, which further improves the reliability of the function of preventing vehicle theft.

[0058] The connector withdrawal prevention portion **201** supports the blade portion **230** so that the blade portion **230** is not folded in a direction in which an angle between the blade portion **230** and the housing coupling portion **210** decreases. For example, in FIG. 8, an angle AN, which is defined as an imaginary straight line L1, which extends in the width direction perpendicular to the longitudinal direction of the housing coupling portion **210**, and an imaginary straight line L2, which extends in the direction in which the blade portion **230** is bent, intersect each other, may be an angle between the blade portion **230** and the housing coupling portion **210**.

[0059] An end **234**, which is opposite to one side end of the blade portion **230** based on the longitudinal direction of the housing coupling portion **210**, overlaps two opposite ends **202** of the connector withdrawal prevention portion **201** based on the width direction of the connector withdrawal prevention portion **201**.

[0060] Therefore, because the two opposite ends **202** of the connector withdrawal prevention portion **201** block and support the blade portion **230** even though an external force is applied in a direction in which the blade portion **230** is folded with respect to the housing coupling portion **210**, the angle AN between the blade portion **230** and the housing coupling portion **210** does not decrease. As a result, the gap between the second inner surface **132** and the blade portion **230** is not enlarged even when an external force is applied by a criminal, which more reliably prevents vehicle theft.

[0061] For example, the anti-theft bracket **200** may be manufactured by performing press processing on a metal board such as stainless steel. However, alternatively, the anti-theft bracket **200** of the present disclosure may be manufactured by injection molding a synthetic resin-based material such as engineering plastic or fiber reinforced plastic. Alternatively, the anti-theft bracket **200** may be formed by casting using a metallic material.

[0062] The present disclosure described above does not require an additional configuration other than the anti-theft bracket **200** to prevent the detachment of the insertion connector **160**. Therefore, the costs required to provide the anti-theft structure of the vehicle may be reduced, and the weight of the vehicle control device **100** may be reduced.

[0063] According to the present disclosure, the anti-theft bracket **200** may be easily coupled to the housing **10** so as not to be separated only by being fitted with the housing **101** without using a fastener such as a bolt. Therefore, it is possible to improve the convenience and productivity of the assembling process.

[0064] While the present disclosure has been described with reference to the exemplary embodiment depicted in the drawings, the exemplary embodiment is described just for illustration, and those skilled in the art to the present technology pertains will understand that various modifications of the exemplary embodiment and any other exemplary embodiment equivalent

thereto are available. Accordingly, the true protection scope of the present disclosure should be determined by the appended claims.

[0065] Although exemplary embodiments of the disclosure have been disclosed for illustrative purposes, those skilled in the art will appreciate that various modifications, additions and substitutions are possible, without departing from the scope and spirit of the disclosure as defined in the accompanying claims. Thus, the true technical scope of the disclosure should be defined by the following claims.

Claims

1. An anti-theft bracket for a vehicle control device, comprising: a connector withdrawal prevention portion configured to block an insertion connector fitted with a housing of the vehicle control device from being withdrawn from the housing; and a housing coupling portion connected to the connector withdrawal prevention portion and fitted with the housing, wherein the housing coupling portion is configured to prevent the housing coupling portion and the connector withdrawal prevention portion from being withdrawn from the housing.
2. The anti-theft bracket of claim 1, wherein the vehicle control device comprises at least one of a body control module (BCM), an integrated body unit (IBU), an engine control module (ECM), or any combination thereof.
3. The anti-theft bracket of claim 1, wherein the housing of the vehicle control device includes a circuit board installed therein, and wherein the insertion connector is electrically coupled to the circuit board.
4. A vehicle control device comprising: a housing comprising: a connector fitting part; and a bracket fitting part provided at one side of the connector fitting part; an insertion connector fitted with the connector fitting part; and an anti-theft bracket comprising: a connector withdrawal prevention portion configured to block the insertion connector fitted with the housing from being withdrawn from the connector fitting part; and a housing coupling portion connected to the connector withdrawal prevention portion and fitted with the bracket fitting part, wherein the housing coupling portion is configured to prevent the housing coupling portion from being withdrawn from the bracket fitting part.
5. The vehicle control device of claim 4, wherein the housing coupling portion comprises a bracket hook protruding in a direction intersecting a direction in which the anti-theft bracket is inserted into the bracket fitting part.
6. The vehicle control device of claim 5, wherein the housing comprises a bracket hook stopper configured to block the bracket hook so that the housing coupling portion inserted into the bracket fitting part does not move in a direction opposite to a direction in which the housing coupling portion is inserted into the bracket fitting part.
7. The vehicle control device of claim 4, wherein the housing comprises a housing hook protruding in a direction intersecting a direction in which the anti-theft bracket is inserted into the bracket fitting part.
8. The vehicle control device of claim 7, wherein the housing coupling portion comprises a housing hook stopper blocked by the housing hook so that the housing coupling portion inserted into the bracket fitting part does not move in a direction opposite to a direction in which the housing coupling portion is inserted into the bracket fitting part.
9. The vehicle control device of claim 4, wherein the connector withdrawal prevention portion covers an inlet of the bracket fitting part.
10. The vehicle control device of claim 4, wherein the anti-theft bracket further comprises a connector insertion portion connected to the connector withdrawal prevention portion and inserted into the insertion connector.
11. The vehicle control device of claim 10, wherein the insertion connector comprises: a groove

portion into which the connector insertion portion is inserted; and a bracket stopper part positioned between the housing coupling portion and the connector insertion portion and facing the connector withdrawal prevention portion when the housing coupling portion is fitted with the bracket fitting part.

12. The vehicle control device of claim 4, wherein the anti-theft bracket further comprises a blade portion being in surface contact with an inner surface that defines an inlet of the bracket fitting part.

13. The vehicle control device of claim 12, wherein the blade portion is bent and extends from the housing coupling portion, and wherein the connector withdrawal prevention portion blocks the blade portion so that the blade portion is not folded in a direction in which an angle between the blade portion and the housing coupling portion decreases.

14. The vehicle control device of claim 4, wherein the anti-theft bracket is manufactured by performing a press processing on a metal board.

15. The vehicle control device of claim 4, wherein the vehicle control device comprises at least one of a body control module (BCM), an integrated body unit (IBU), an engine control module (ECM), or any combination thereof.

16. The anti-theft bracket of claim 4, wherein the housing of the vehicle control device includes a circuit board installed therein, and wherein the insertion connector is electrically coupled to the circuit board.
